Cambridge (CIE) A Level Computer Science

Your notes

Logic Gates & Logic Circuits

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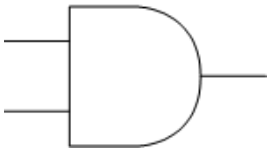
Your notes

Logic gates

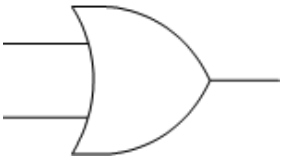
What is a logic gate?

- A logic gate is a **visual way** of representing a **Boolean expression**
- The logic gates covered in this course are:
  - **AND**
  - **OR**
  - **NOT**
  - **XOR**
  - **NAND**
  - **NOR**

AND

Expression operator	Circuit symbol	Function
(A **AND** B)		Returns TRUE only if **both** inputs are TRUE **TRUE** AND **TRUE** = **TRUE** Otherwise = FALSE

OR

Expression operator	Circuit symbol	Function
(A **OR** B)		Returns TRUE if **either** input is TRUE **TRUE** OR **FALSE** = **TRUE** **FALSE** OR **FALSE** = **FALSE**

NOT

Expression operator	Circuit symbol	Function

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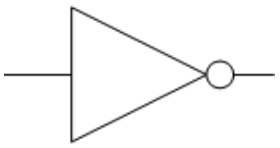
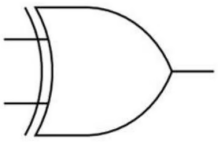
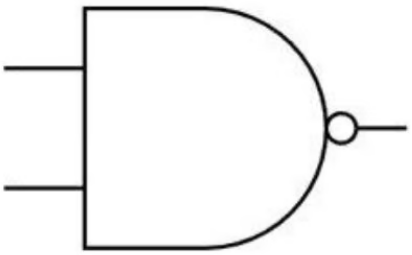
(**NOT** A)		**Reverses** the input value NOT **TRUE** = **FALSE** NOT **FALSE** = **TRUE**

Your notes

XOR (exclusive OR)

Expression operator	Circuit symbol	Function
(A **XOR** B)		Returns TRUE if **either** input is TRUE but **NOT** both **TRUE** OR **FALSE** = **TRUE** **FALSE** OR **FALSE** = **FALSE** **TRUE** OR **TRUE** = **FALSE**

NAND (not and)

Expression operator	Circuit symbol	Function
(A **NAND** B)		Returns TRUE if **either or both** inputs are FALSE **TRUE** OR **FALSE** = **TRUE** **FALSE** OR **FALSE** = **TRUE** **TRUE** OR **TRUE** = **FALSE**

NOR (not OR)

Expression operator	Circuit symbol	Function

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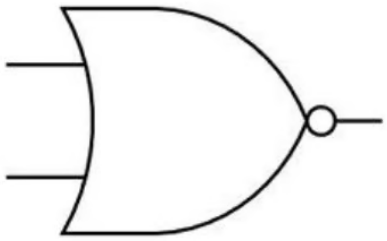
(A **NOR** B)		Returns TRUE if **both** inputs are FALSE **TRUE** OR **FALSE** = **FALSE** **FALSE** OR **FALSE** = **TRUE** **TRUE** OR **TRUE** = **FALSE**

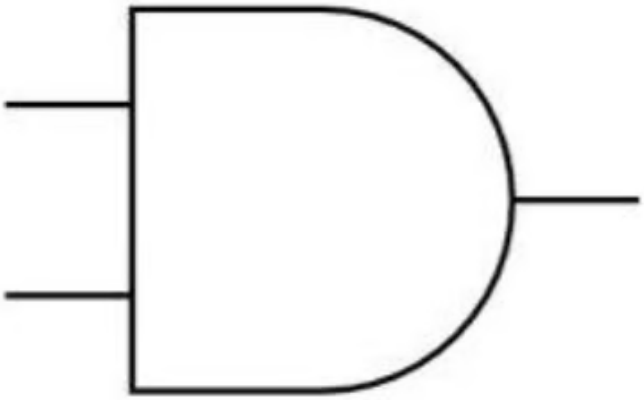
Your notes

Truth tables

What is a truth table?

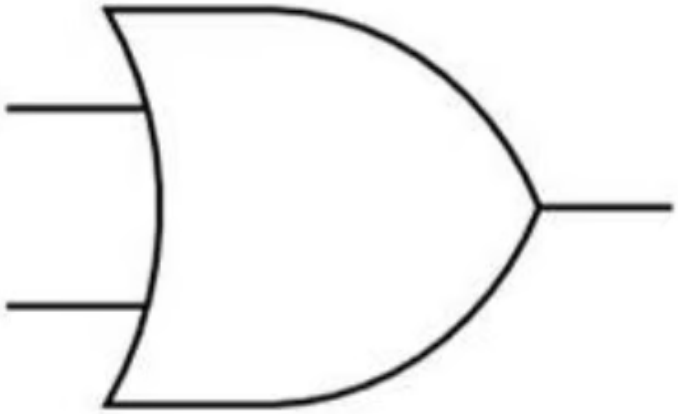
- A truth table is a **tool** used in logic and computer science to **visualise** the **results** of **Boolean expressions**
- They represent **all possible inputs** and the **associated outputs** for a **given Boolean expression**

AND

Circuit symbol	Truth Table
	

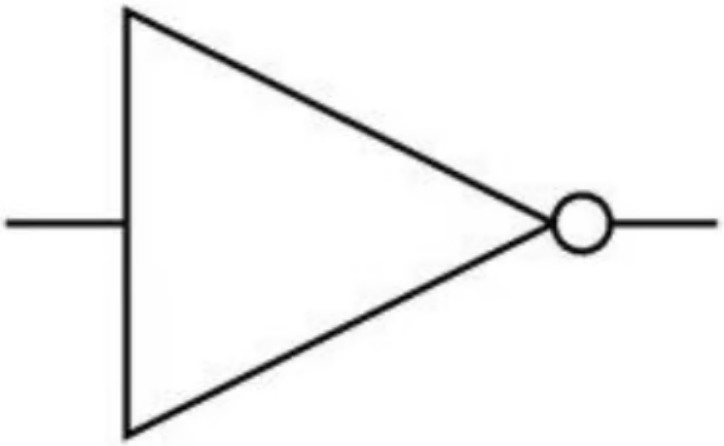
A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

OR

Circuit symbol	Truth Table
	

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

NOT

Circuit symbol	Truth Table
	

A	NOT
0	1
1	0

XOR (exclusive)

Your notes

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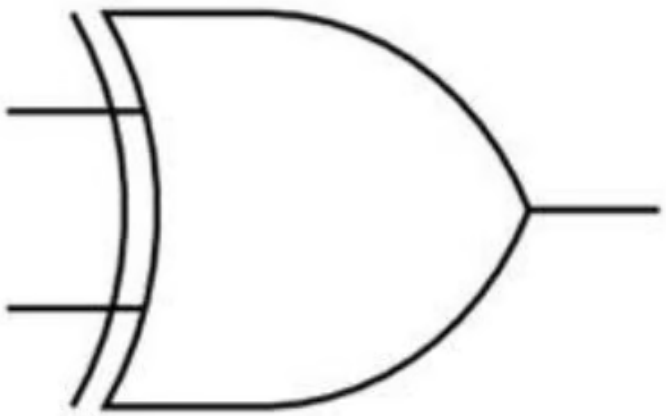
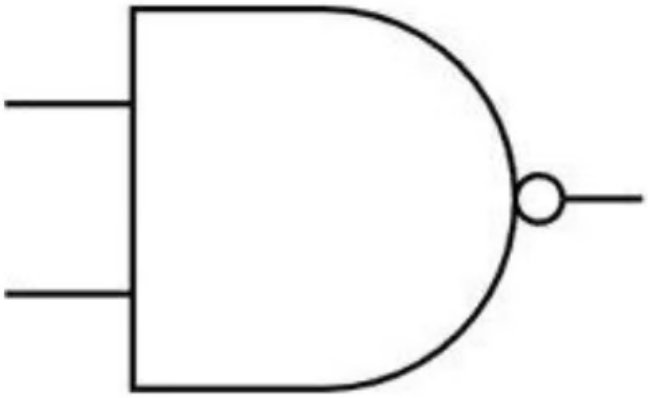
Circuit symbol	Truth Table
	

A	B	A XOR B
0	0	0
0	1	1
1	0	1
1	1	0

NAND (not and)

Circuit symbol	Truth Table
	

A	B	NOT (A AND B)
0	0	1
0	1	1
1	0	1
1	1	0

NOR (not or)

Circuit symbol	Truth Table

Your notes

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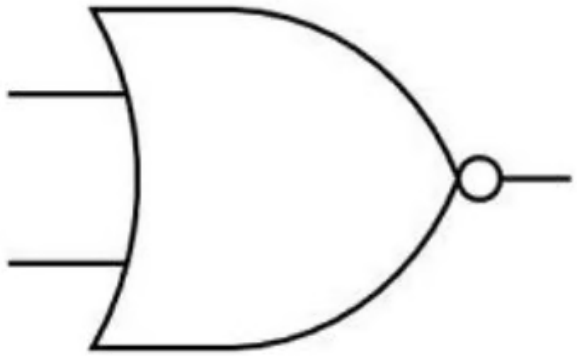


A	B	NOT (A OR B)
0	0	1
0	1	0
1	0	0
1	1	0

Your notes

Your notes

From problem statements

- A problem statement is typically a **written interpretation of a scenario** that requires a **specific logical outcome**
- Logic circuits can be **constructed using from problem statements**

Worked Example

A server room has an **automatic cooling fan (F)**
The fan turns on based on the following input parameters:

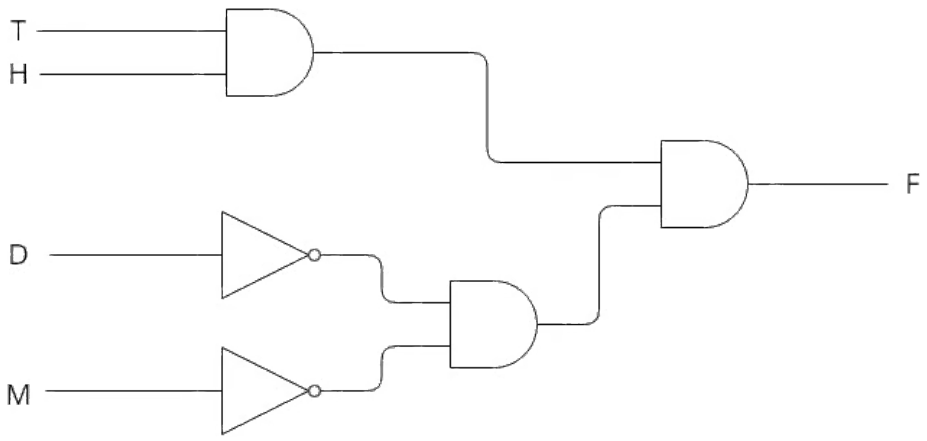
Parameter	Description	Binary Value	Condition
T	Temperature	1 = Too high	0 = Acceptable
H	Humidity	1 = Too high	0 = Acceptable
M	Maintenance mode	1 = Active	0 = Inactive
D	Door sensor	1 = Door open	0 = Door closed

The **fan (F = 1)** turns on if:

- The **temperature is too high**
- **AND** humidity is too high
- **AND** the **door is closed**
- **UNLESS** maintenance mode is active (if maintenance is active, the fan must stay off)

Draw a logic circuit to represent how the **fan (F)** operates based on the input conditions. **[3]**

Answer



- **NOT gates** on **D** and **M** inputs [1 mark]
- AND gate combines **T AND H** or **NOT D AND NOT M** [1 mark]
- Combines two previous AND outputs (e.g. (**T AND H) AND (NOT D AND NOT M)**) [1 mark]

Your notes

From logic expressions

- A logic expression is a way of showing how a **logic circuit works using symbols** and **Boolean logic** (AND, OR, NOT)
- It describes the conditions that must be met for the **output** to be **true (1)** or **false (0)**
- From a logic expression, a **logic circuit and/or truth table can be constructed**

Worked Example

A logic expression is given:

S = (A AND B AND C) OR (B XOR C)

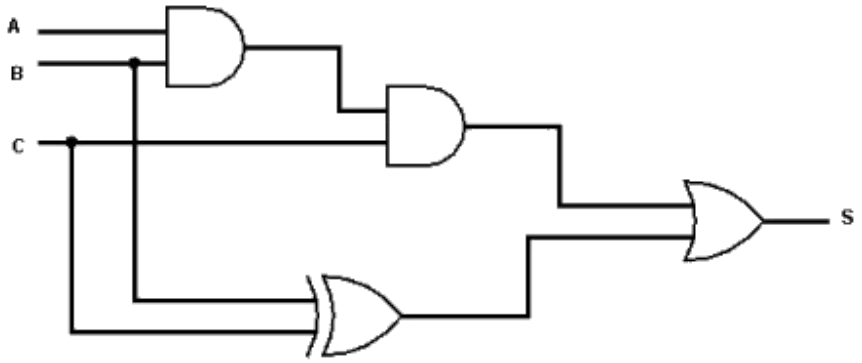
(a) Draw the logic circuit for the given expression **[4]**

(b) Complete the truth table for the logic expression: **[2]**

S = (A AND B AND C) OR (B XOR C)

Answer

(a)



- Each correct gate [1 mark]

(b)

A	B	C	S
0	0	0	0
0	0	1	1

0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

- Rows **1 to 4** correct [1 mark]
- Rows **5 to 8** correct [1 mark]

Your notes

From truth tables

- To create a truth table for the expression **P = (A AND B) AND NOT C**
  - Calculate the numbers of rows needed ($2^{\text{number of inputs}}$)
  - In this example there are 3 inputs (**A, B, C**) so a total of **8 rows** are needed (2^3)
  - To not miss any combination of inputs, start with **000** and **count** up in **3-bit binary (0-7)**

A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

- Add a new column to show the **results** of the brackets first (A AND B)

A	B	C	A AND B

0	0	0	**0**
0	0	1	**0**
0	1	0	**0**
0	1	1	**0**
1	0	0	**0**
1	0	1	**0**
1	1	0	**1**
1	1	1	**1**

- Add a new column to show the **results** of NOT C

A	B	C	A AND B	NOT C
0	0	0	0	**1**
0	0	1	0	**0**
0	1	0	0	**1**
0	1	1	0	**0**
1	0	0	0	**1**
1	0	1	0	**0**
1	1	0	1	**1**
1	1	1	1	**0**

- The last column shows the **result** of the Boolean expression (**P**) by comparing (A AND B) AND NOT C //

A	B	C	A AND B	NOT C	P
0	0	0	0	1	**0**

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0	0	1	0	0	**0**
0	1	0	0	1	**0**
0	1	1	0	0	**0**
1	0	0	0	1	**0**
1	0	1	0	0	**0**
1	1	0	1	1	**1**
1	1	1	1	0	**0**

Your notes

Examiner Tips and Tricks

It is possible to create a truth table when combining expressions that show only the inputs and the final outputs.

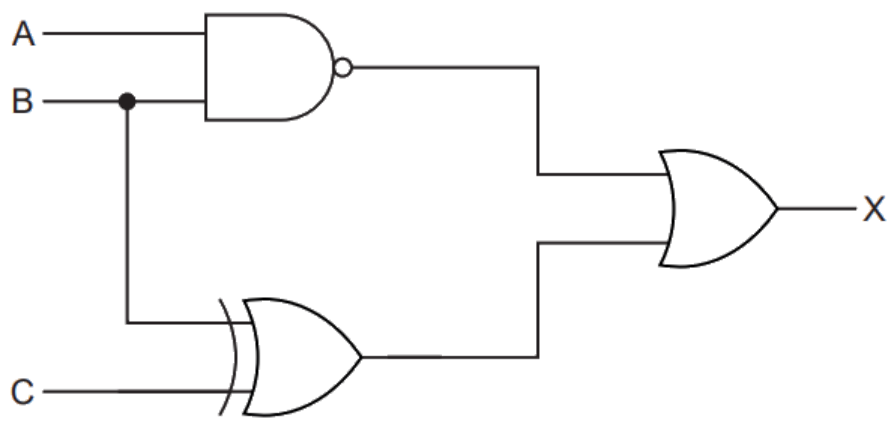
The inclusion of the extra columns **supports the process** but can be skipped if you feel able to do those in your head as you go.

From logic circuits

- From a logic circuit you can create the **logic expression and/or truth table**

Worked Example

Write the logic expression for the given logic circuit. **[3]**



Answer

(A NAND B) OR (B XOR C)

- A NAND B [1 mark]
- B XOR C [1 mark]
- OR [1 mark]

Your notes